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Steganographic Algorithm Injection in Image Information Systems used in Healthcare Organizations

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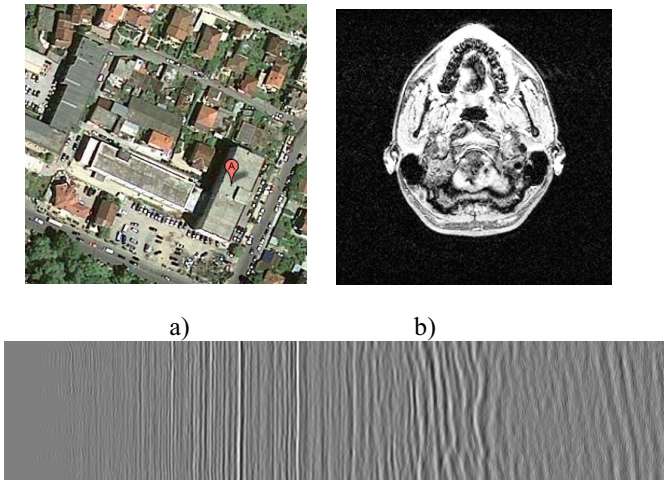
***Abstract*— Image Information Systems, as every other information system where we store digital data, is frequently attacked by outer intruders who want to possess these data and use them to their interest. In order to fight these phenomena information systems built on databases use different security techniques. Nevertheless, at the moment that system data, in our case images, are exposed on the web, they cannot be considered protected, since they after the copy process do not show clearly their author. In this paper we offer a secure way regarding image authenticity. Steganography is the science of hiding data in electronic images and we propose a steganographic algorithm that injects image metadata in the image itself. These metadata are not only regarding the image itself, but an important part of them is also an ID generated according to our scheme. We will focus on IIS of Healthcare organizations. For this reason we will try to guaranty authenticity without touching the image quality. The proposed injection is tested in AMP systems and it is done prior the image is stored in the database.**

***Keywords-component;* Steganography; Image Information Systems; authenticity; algorithm injection**

I. INTRODUCTION

Image Information Systems are a combination of Information Technology and other human activities that use this technology for different purposes, to manage, to make decisions, to simply store or retrieve information or to process data. Specifically, they capture, store, analyze, manage and display images of a given organization. They are used widely in the most important institutions, since images are one of the most important media for important business companies, or scientific institutions. In a certain manner, these systems are a global representation of what we are more familiar with, such as Geographical Information Systems, Multimedia Systems, etc. There are many research fields regarding Image Information Systems (IISs) such as progressive representation of unstructured data, investigation of data mining techniques, security issues, and many others. These systems are very necessary especially when we deal with large amount of data. Massive data stored, analyzed or processed in IISs, are usually taken from natural data such as satellite images, seismic images or even human body images such as those taken in health centers.

In our paper we will focus on one of the most important aspects of IISs: data security. We will introduce a technique to better protect images from intruders in the system. In order to fight these phenomena information systems use different security techniques which are offered by the platforms on which their databases are built. Nevertheless, at the moment that system data, in our case images, are exposed on the web, they cannot be considered protected, since they after the copy process do not show clearly their author.



a) b)

c)

Figure 1. Examples of images stored in IIS. a) satellite view, Faculty of Economy, Tirana, b) medical image, scaner, c) seismic data

In this paper we also offer a secure way regarding image authenticity. Our scheme guaranties authenticity without touching the image quality [1], [2]. The metadata inserted are not only regarding the image itself, but the most important part of them is an ID number. We will focus on IIS of Healthcare organizations. For this reason we will try to guaranty authenticity without touching the image quality.

We are unpleased to mention that IISs are not used very much in the Balkans. From our study in the three countries of the region (Albania, Macedonia and Kosovo), there are only a few research projects regarding GIS only. Other

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similar systems used by our institutions are mostly data base systems and do not offer other services. This is a greater motivation for us to try to bring to the front the importance of these systems and try also to improve their data security.

II. STEGANOGRAPHIC PROTECTION OF IMAGES

Trying to protect the information of a system means to not allow the unauthorized intrusion, usage, change or destruction of its data. This protection is not only a business, ethical or legal request but sometimes even vital to the system itself. If we consider for example marketing or advertising agencies, or graphical studios, it is quite clear that author rights of the images produced by them is vital to the system. Some of the main points to be considered are:

1. Communication of data
2. Attacks of intruders in the system
3. Digital protection

One of the safest ways to prove the authenticity of an image is to use steganographic techniques. Steganography is the field of study that tries to keep hidden the message intended to be communicated between a source and a destination, or between Alice and Bob [3]. Figure 2 explains better the realization of a steganographic technique. The carrier image serves as a container of binary data. In order to make a safer communication, text is coded by a key. The data is inserted one by one in every pixel of the carrier image, which now is called stegoimage.

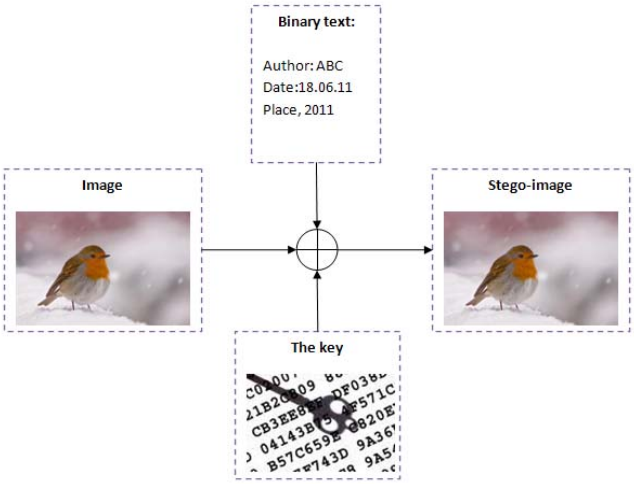

Figure 2. Schematic representation of a steganographic technique

III. STEGIM - THE ALGORITHM INJECTED IN IISS

The algorithm we are going to introduce in this section can be applied in Image Information Systems in order to show in a hidden manner the property of it. By doing so it contributed significantly in the author rights protection, but if used on IISs applied in Healthcare organizations it also contributes in storing the privacy of patients.

Medical images come from a board spectrum of imaging sources such as:

- computed axial tomography (CT)
- magnetic resonance imaging (MRI)
- digital mammography
- positron mission tomography (PET)

They generate a staggering amount of image data and important medical information [2]. Thus, the most essential data for an image might be:

1. Patient name
2. ID of image (from our generation scheme)
3. Type of image source (CT, MRI, PET)
4. Name of Healthcare Organization
5. Date

This data is all converted to binary and inserted into the image pixels. Prior the moment of storing the image in the database, it must pass through our algorithm – StegIm.

Also, the whole text will be repeated in the ROI part of the image. The text will be encoded by using the AES technique. The unique code of the image will also serve as a watermark for the image. It will be stamped on the left upper corner of the image.

The reverse process, which is the identification and content extraction from the image can be done by reading the LSB of every pixel, creating a chain of bits and group them into 8, converting the group from the ASCII table of characters and having so the full text.

*StegIM Algorithm: * Encryption Process*

```
Begin
Read Image from source
Read Message from User
Read Message from the System
msg_id=GenerateID
a = msg_s + msg_p + ID
GenerateKEY()
msg_a = AESencryption (a)
      //we obtain msg_a from the encryption of a
msg_b = BinaryConvertion (msg_a)
      //we obtain msg_b  from the convertion of msg_a
For I = 20  to N
      For J = 1 to M
                If lastElement(msg_b) = NULL
                          startOver (msg_b)
                Else
      LSBbit (T[I, J]) = lastElement (msg_b)
      nextElement(msg_b)
      EndFor
EndFor                              //stego-image stored
Open DB
Query Execution
Store Image in  DB
End
```

Figure 3 shows the ROI part of a medical image. It is the lighter grey part and it is not visible to the human eye. The proposed metadata will be injected in every row according to the ROI scheme.

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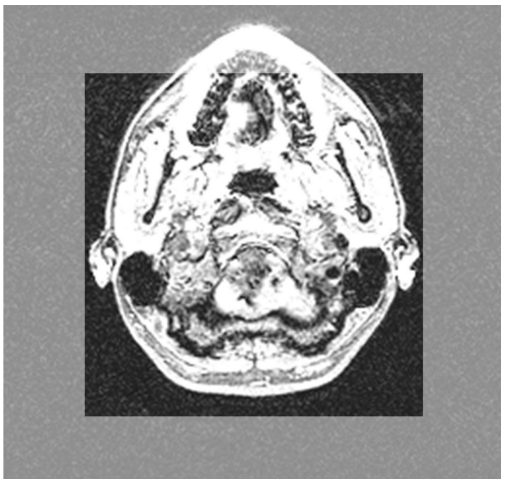

Figure 3. Region Of Interest of a medical image

Let us supposea patient with the following data:
Patient: John Lee
ID: 89903158929341000
Type: MRI
Name: German Hospital Tirana
Date 01.09.2011

The data is converted to binary and inserted as a LSB of the image. The following bits result from this conversion.

01001010 01101111 01101000 01101110 00100000
01001100 01100101 01100101 00001101 00001010
00111000 00111001 00111001 00110000 00110011
00110001 00110101 00111000 00111001 00110010
00111001 00110011 00110100 00110001 00110000
00110000 00110000 00001101 00001010 01001101
01010010 01001001 00001101 00001010 01000111
01100101 01110010 01101101 01100001 01101110
00100000 01001000 01101111 01110011 01110000
01101001 01110100 01100001 01101100 00100000
01010100 01101001 01110010 01100001 01101110
01100001 00001101 00001010 00110000 00110001
00101110 00110000 00111001 00101110 00110010
00110000 00110001 00110001

A. Generation Scheme of ID numberof patients

We have built an ID generation scheme in order to verify that a certain image is unique and belongs to the Healthcare organization and to a proper patient. The scheme which is a monotone digital signature is an adoption of Rabin digital signature scheme. [6], [7]. It identifies false images. It also offers a decentralized verification system. The ID generated is not dependent on the data itself. This scheme was first asked by the Agency for Achievement Evaluation in the Education Directorate of Tirana, Albania. They needed a security scheme in their certificates of high school students. This system must identify documents (being them diplomas, images, or other electronic document) and must be resistant toward falsifications. It also should offer a decentralized verification system. The system we are going to introduce presumes that verification of patient data (images) can be done by local educational directorates of the Healthcare Organization.

The codes are stamped on the lower part of the Region of Interest of the image (see fig. 3). It is based on a secret key which belongs to the organization. Our idea is that after the generation of n-couples of keys {(si, vi)}i=1..n, the scheme keeps hidden the half {(si)}i=1..n (compounded by the first coordinates) of it and makes public the other part of it {(vi)}i=1..n, for any j<n (compounded by the second coordinates).

If the scheme generator is obliged or forced to give any information regarding the scheme, he can show to the falsifiers who own the signature algorithm and also the secret keys {si}i=1..j, needed to build a real subscription. With all these a falsifier would build "real" signatures which would not be perceived as such at level j of control. At this moment, if the signature holder is released the central directorate sends to all local directorates the level j+1 of security, with which they must check all signatures generated till that moment. False signatures did not consider (since they did not know) the keys sj+1, ..., sn, and as a result $V_{j+1}(v_1,v_2,...,v_{j+1}, m, \sigma) \neq V$. This way, it is easy to see that the document is false. Patient data are very sensitive and it is very important to preserve their secrecy. Also one of the advantages of the inserted data into the body of the image strongly reduces the risk of data confusion between patients.

IV. Conclusions

In this paper we offered a way of protecting the author's right of digital images. We considered the case of Image Information Systems used in Healthcare organizations and proposed an algorithm which injects binary content in the image itself. This is done by means of steganographic techniques.

In this paper we introduced how to use an improved Rabin signature to increase the security level. We suggest implementing this scheme in IISs of Healthcare organizations prior the storage of the images. One of the advantages of this scheme is that it is resistant by a high degree to key generation software. Even though they may detect the code, they will do so only temporarily since the scheme changes the codes over time.

We hope to implement our techniques in a real IIS. They can also be adopted in other Information Systems such as Multimedia Information Systems. The amount of information stored in videos is much more than in images, so the gain would be higher.

Lastly, we must mention that the steganographic technique is vulnerable from image processing, but researchers are working intensively to solve this issue.

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